



Alameed university-Faculty of medicine



Histology of the Circulatory System L2

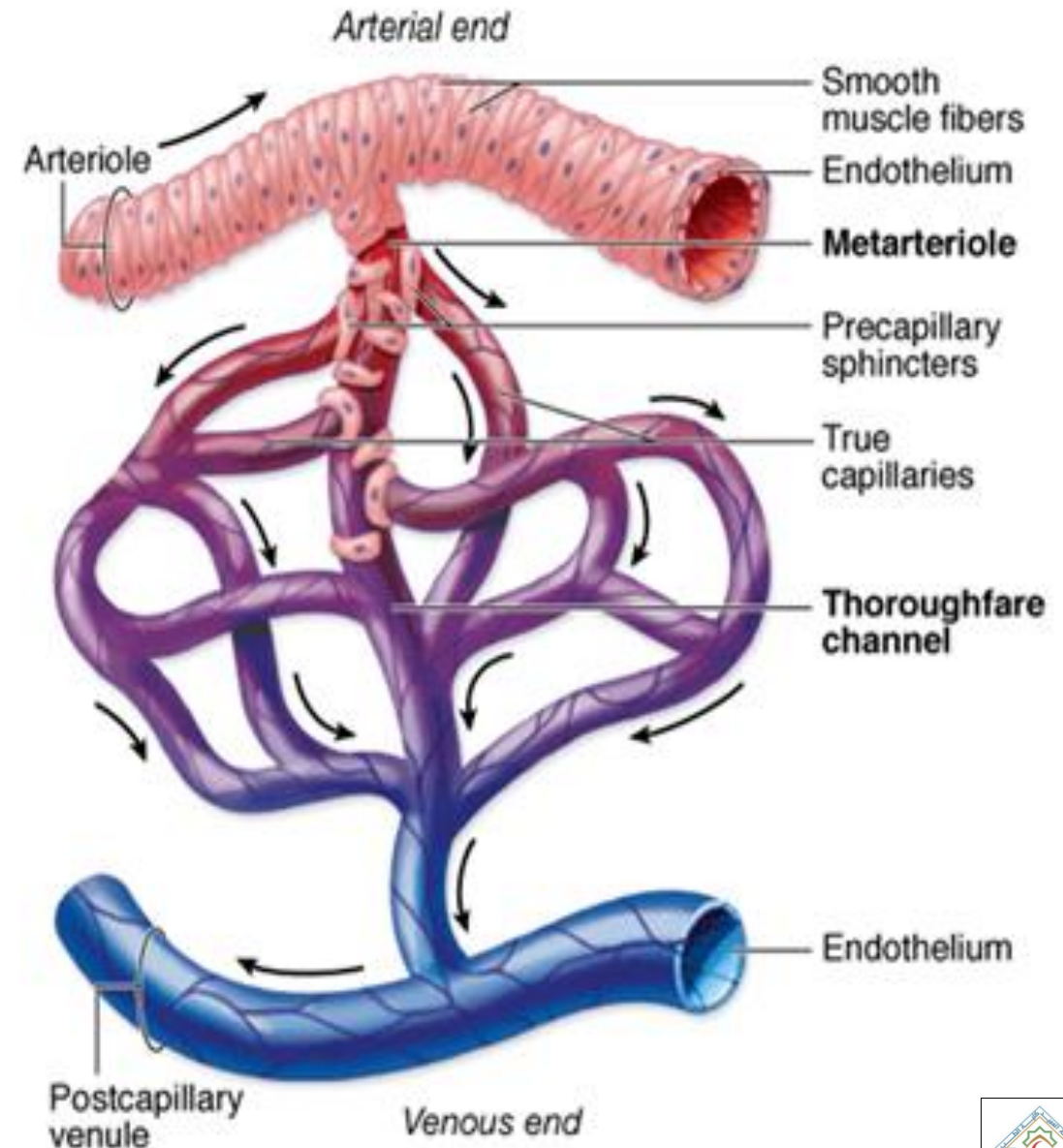
Dr. Shehab Ahmad Faris

Capillaries permit and regulate metabolic exchange between blood and surrounding tissues. These smallest blood vessels always function in networks called capillary beds.

1. Capillary beds are supplied by one or more terminal arteriole branches called **metarterioles**, which are continuous with **thoroughfare channels** connected with the **postcapillary venules**.

2. Capillaries branch from the metarterioles, which are encircled by smooth muscle cells, act as **precapillary sphincters** that control blood flow into the capillaries that converge into the thoroughfare channels, which lack muscle.

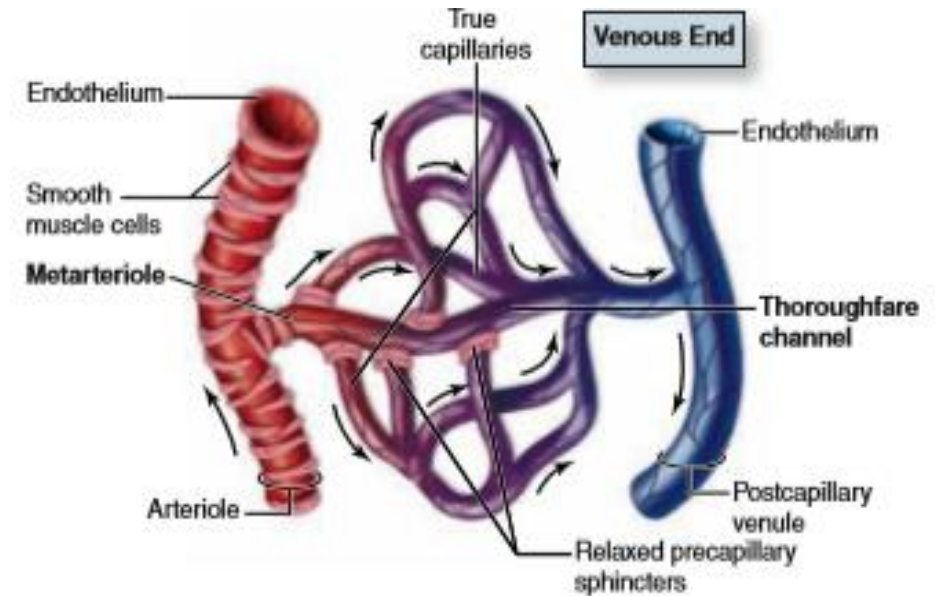
Capillary Beds



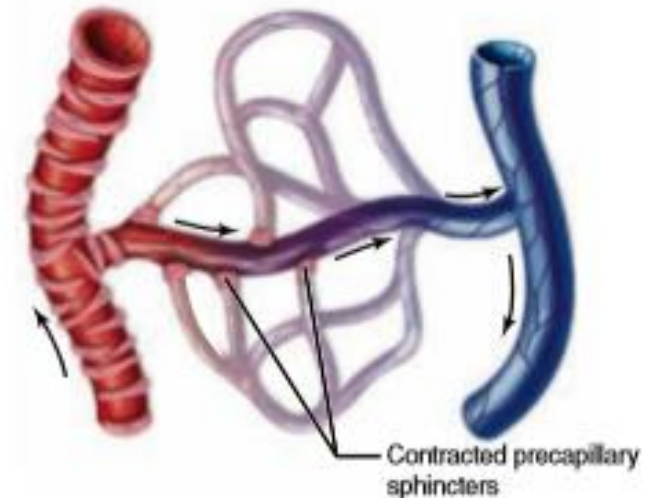
Capillary Beds

3. These sphincters contract and relax cyclically, with 5-10 cycles per minute, causing blood to pass through capillaries in a pulsatile manner.

4. When the sphincters are **relaxed**, capillary bed well **perfused**. When the sphincters are closed, blood flows directly from the **metarterioles and thoroughfare channels** into postcapillary venules.



(a) Sphincters relaxed; capillary bed well perfused



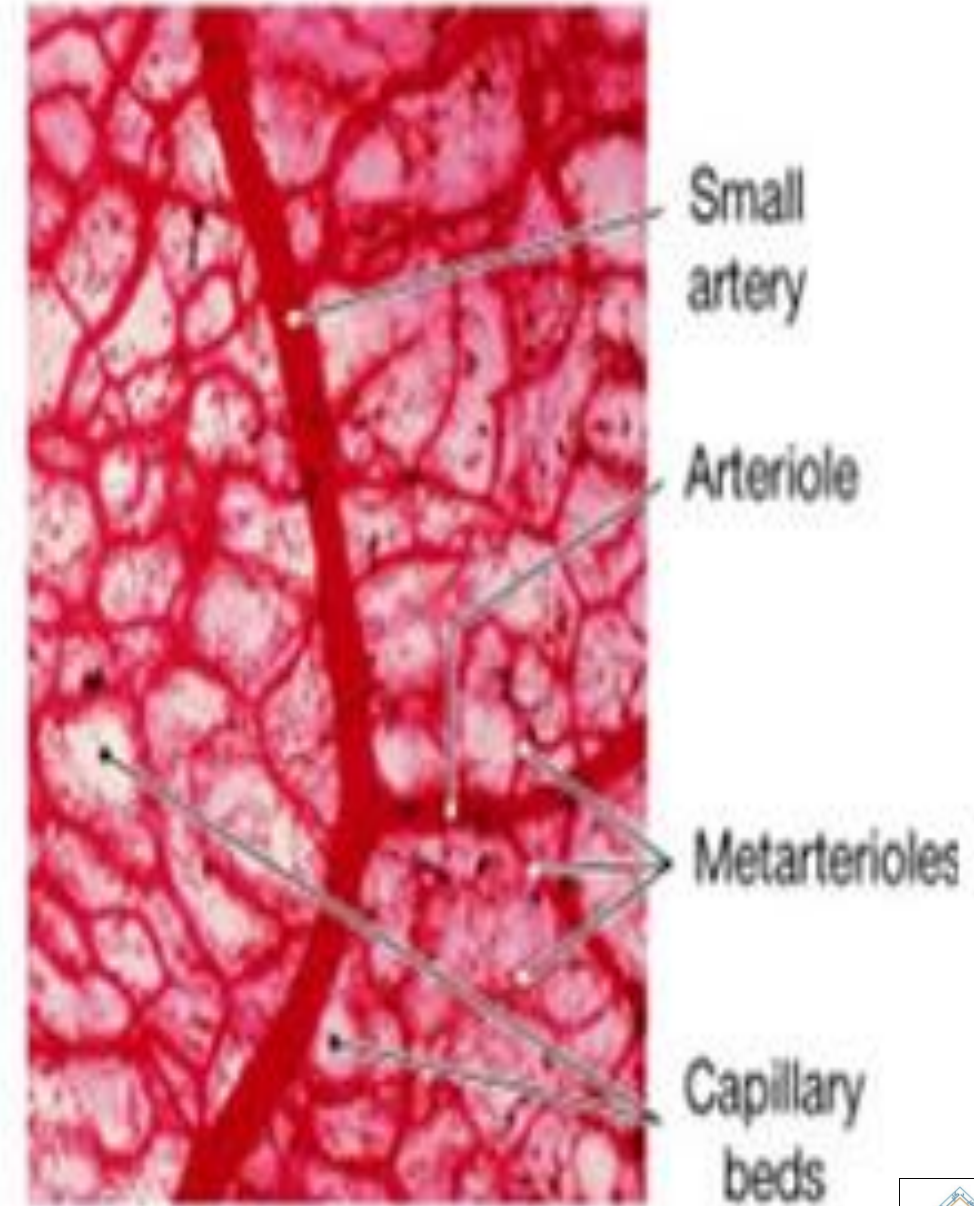
(b) Sphincters contracted; blood bypasses capillary bed



Capillary Beds

5. Tissues with high metabolic rates, such as the **kidney, liver, and cardiac and skeletal muscle**, have an abundant capillary network

6. The opposite is true of tissues with low metabolic rates, such as **smooth muscle and dense connective tissue**.

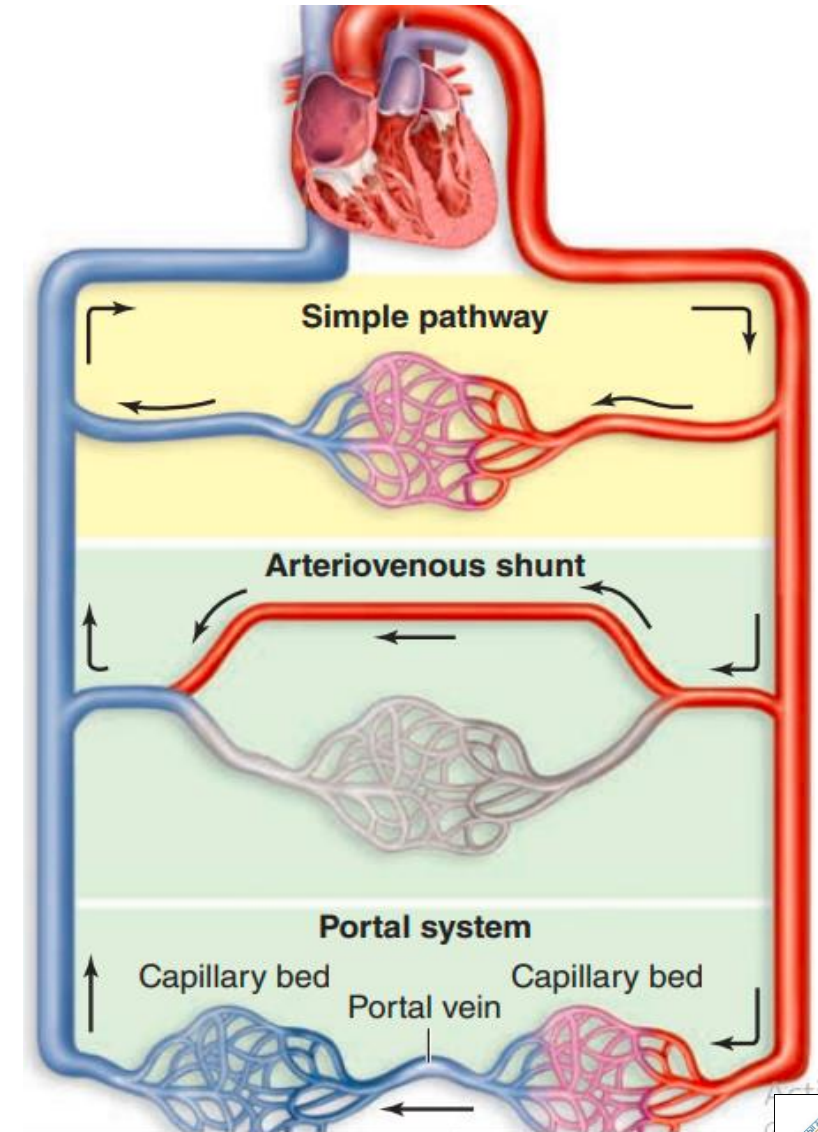


Arteriovenous (AV) shunts, or anastomoses

Most capillary beds are supplied by arterioles and drain into venules. Arteriovenous (AV) shunts are alternative pathways found in certain organs such as skin.

Arteriovenous (AV) shunts which directly connect the arterial and venous systems and temporarily bypass capillaries.

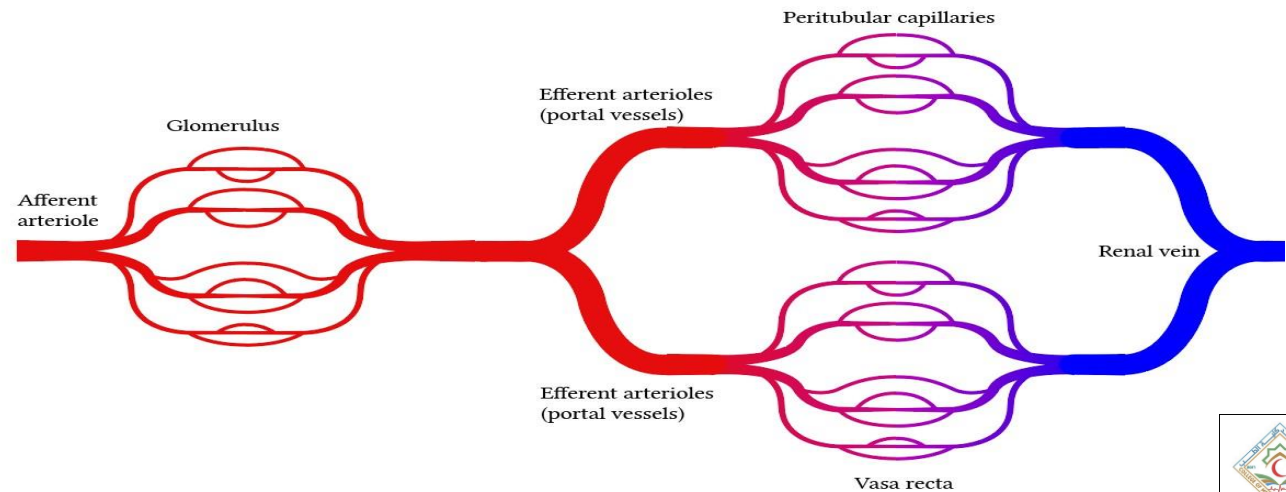
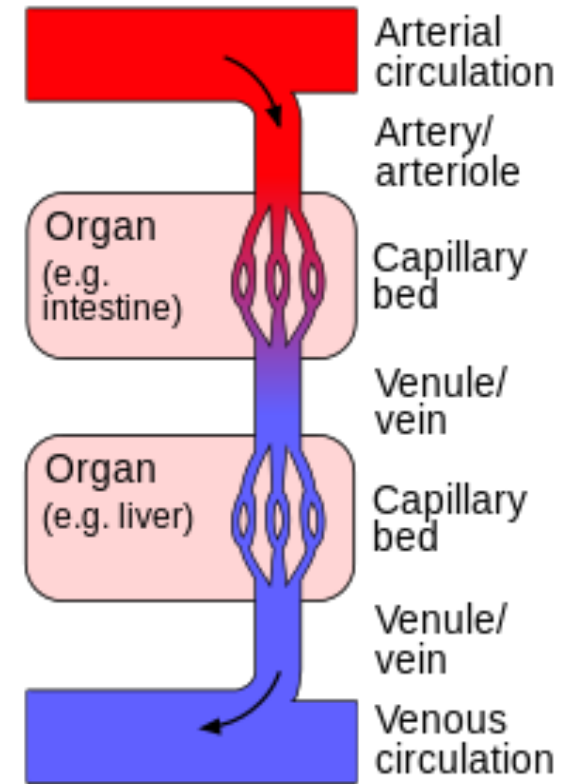
In skin blood flow can be varied according to external conditions.



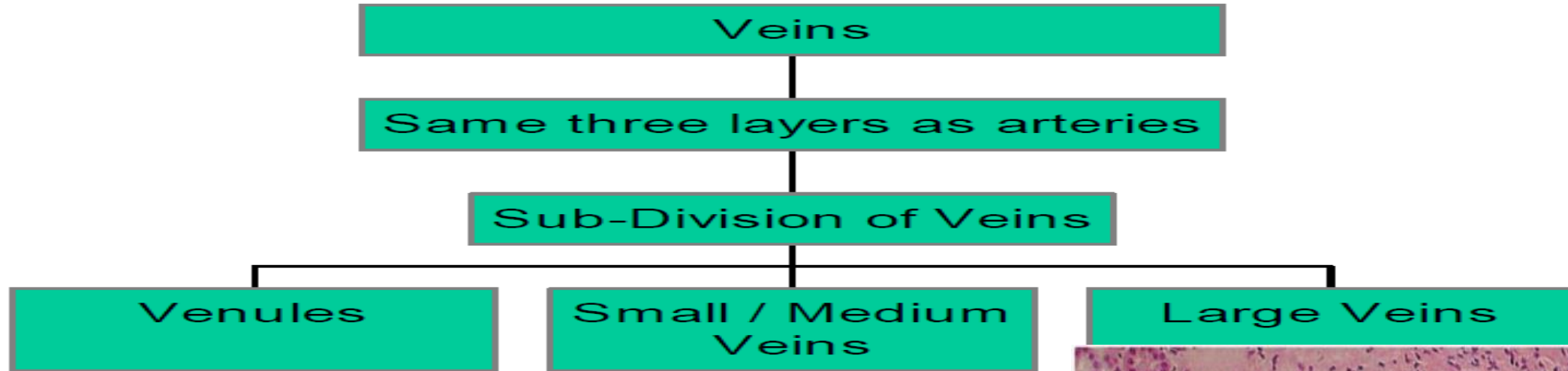
Portal systems

1. In venous portal systems one capillary bed drains into a vein that then branches again into another capillary bed. which is important in the anterior **pituitary gland** and **liver**.

2. Arterial portal systems (afferent arteriole → capillaries → efferent arteriole) which occur in **the kidney**



C. Veins



Thin wall ,wide lumen, thin media, poorly developed smooth muscles and minimum elastic fibers. Veins carry blood back to the heart from microvasculature under very low pressure



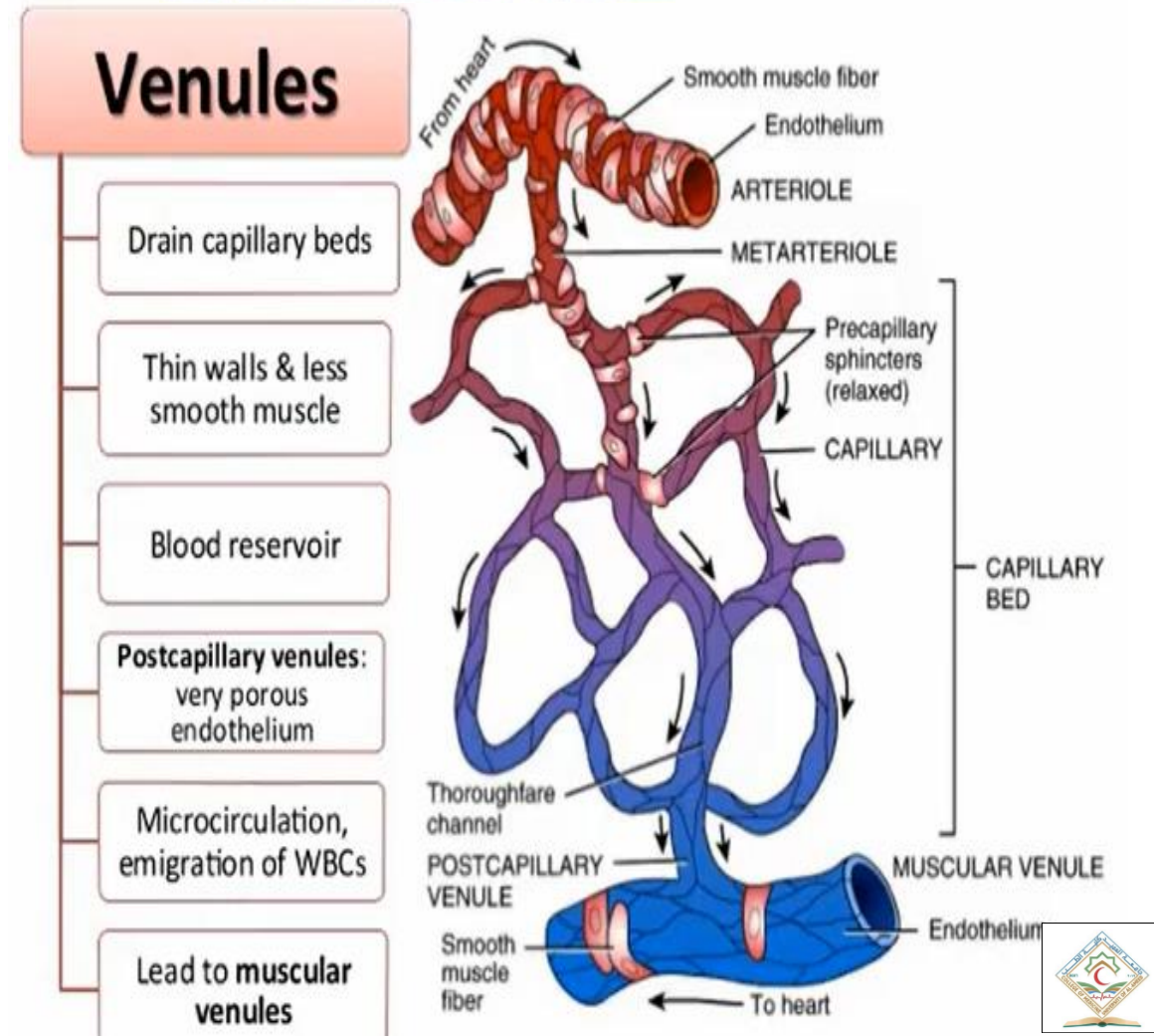
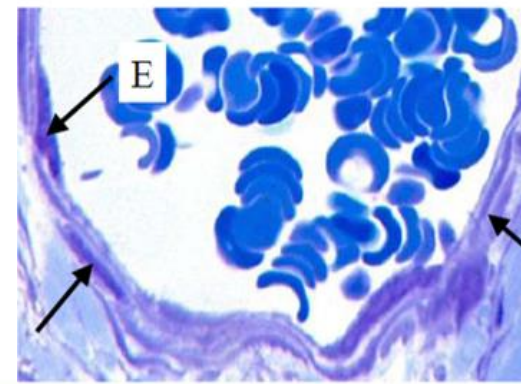
1. Venules

Classified into

a. Postcapillary venules. Single layer of endothelial cells resting on basal lamina surrounded by pericytes.

Junctions between endothelial cells of postcapillary venules are the loosest of the microvasculature. This facilitates transendothelial **migration of leukocytes** at these locations during inflammation, as well as a characteristic loss of fluid here during the inflammatory response, leading to **tissue edema**

b. Muscular venules. The media is formed of **1 to 2 smooth muscle cells** and thin adventitia



2. Small and medium-sized veins

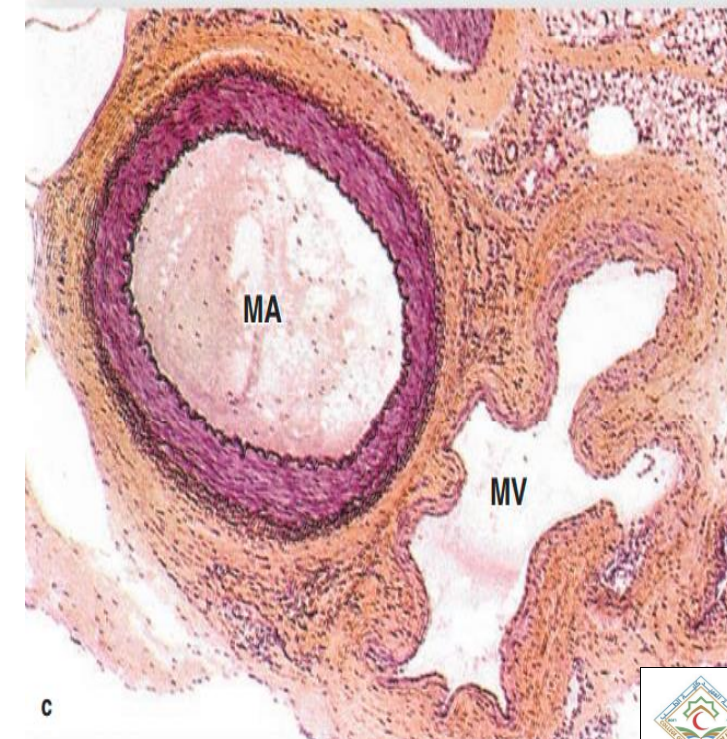
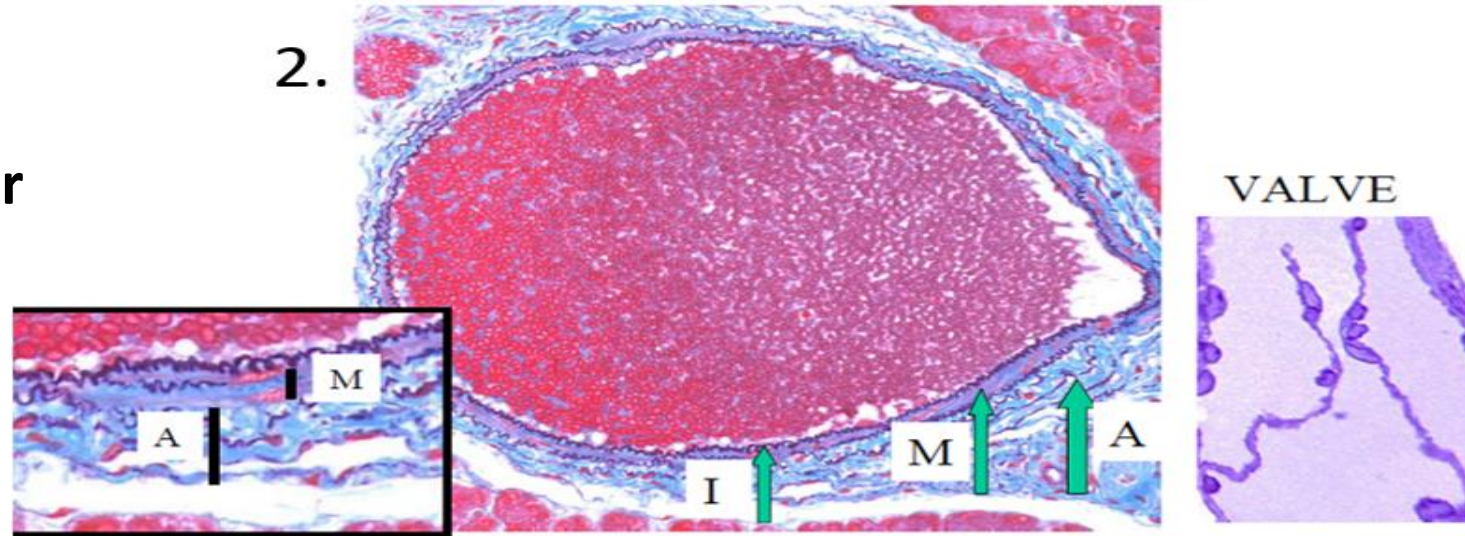
These veins are usually located close and parallel to corresponding muscular arteries.

- a. The tunica intima is usually thin, containing sub endothelial tissue
- b. The media has small bundles of smooth muscle cells mixed with a network of reticular fibers and delicate elastic fibers.
- c. The adventitia is thicker than media contains collagen fibers with small amount of smooth muscles and elastic fibers.

Include most named veins with diameters of 10 mm or less. Vasa vasorum are numerous

Small & Medium - Sized Veins

2.



3. Large Veins

Include vena cava, subclavian and portal veins.

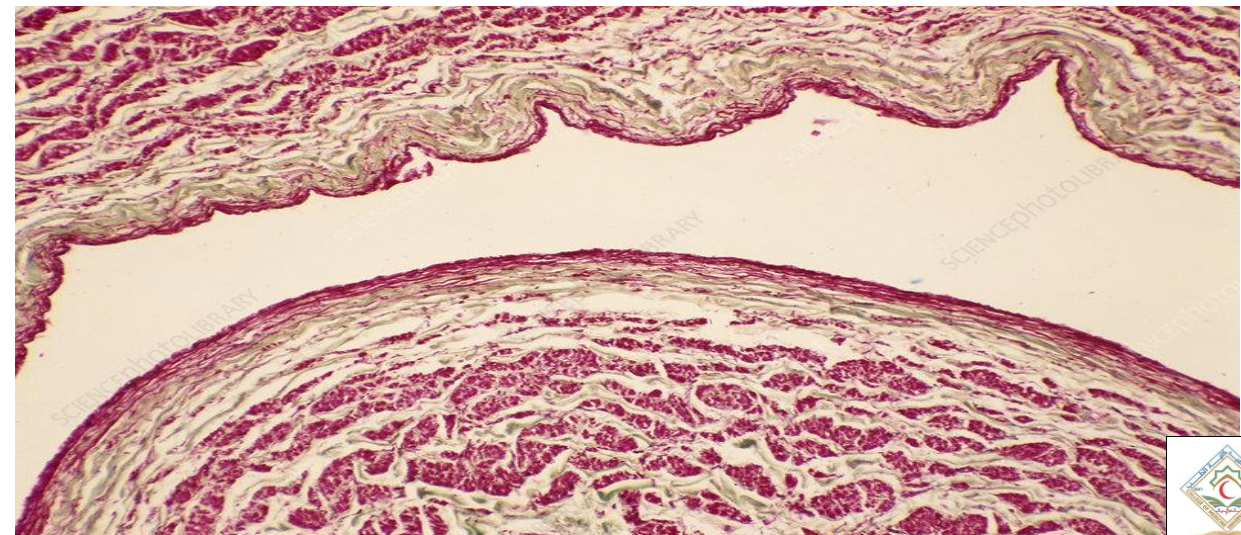
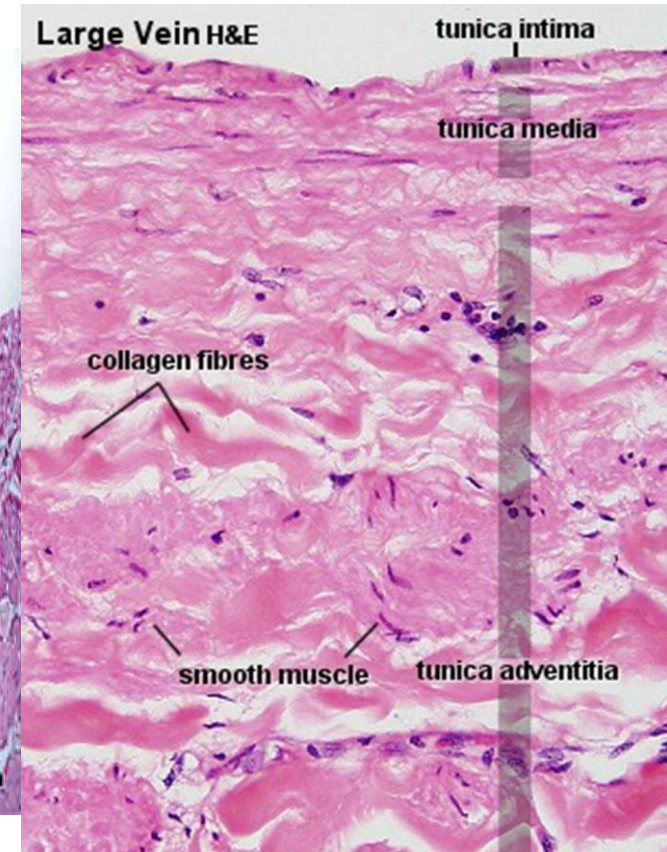
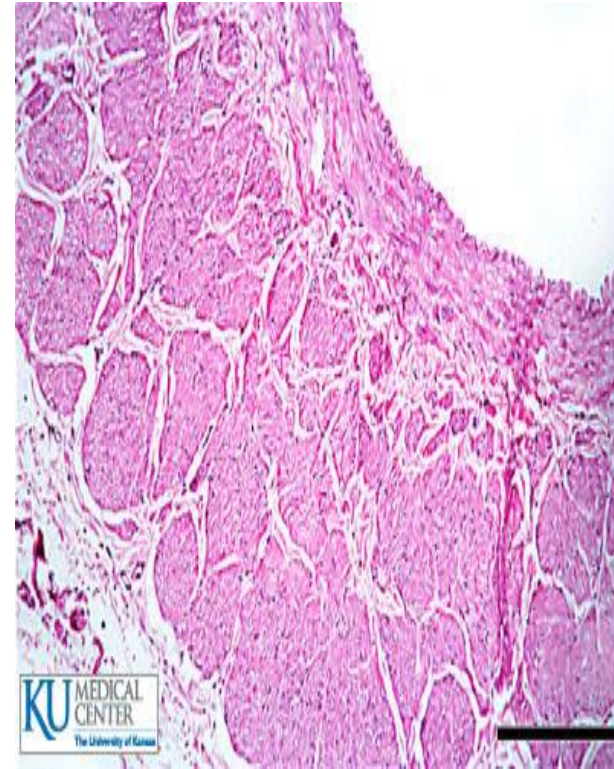
General structure of blood vessels and...

a. Tunica intima with no internal elastic lamina.

b. Relatively thin media 30% with alternating concentric smooth muscle and collagen fibers. no external elastic lamina.

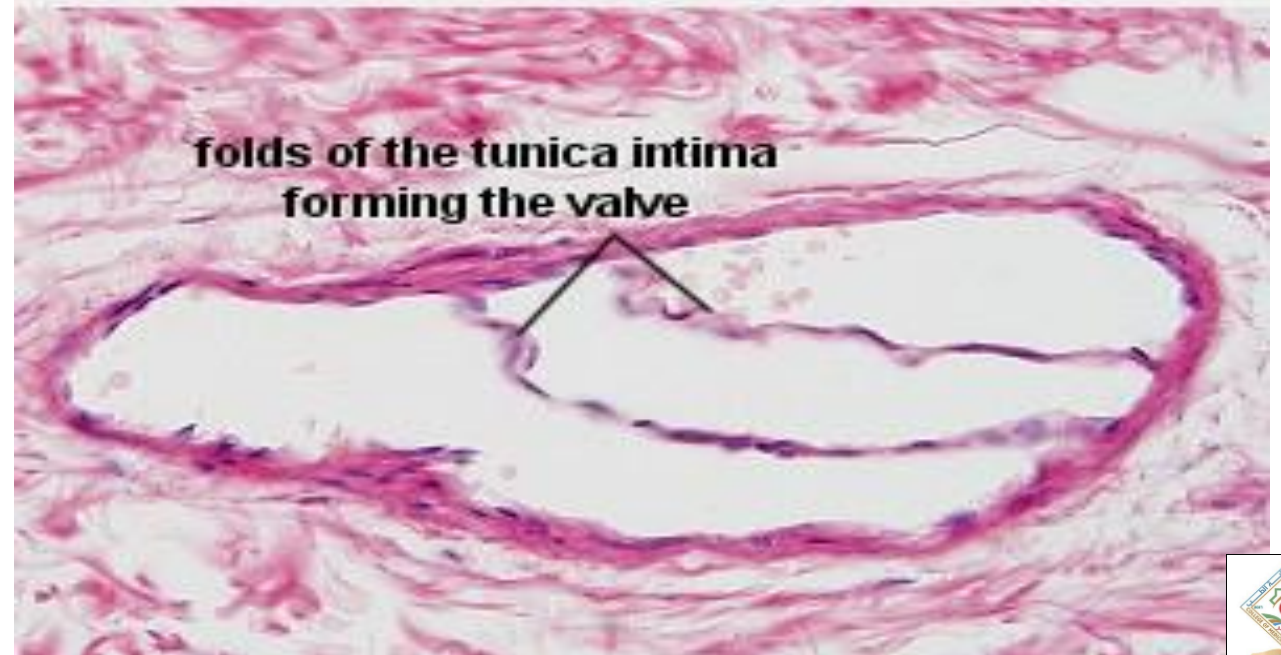
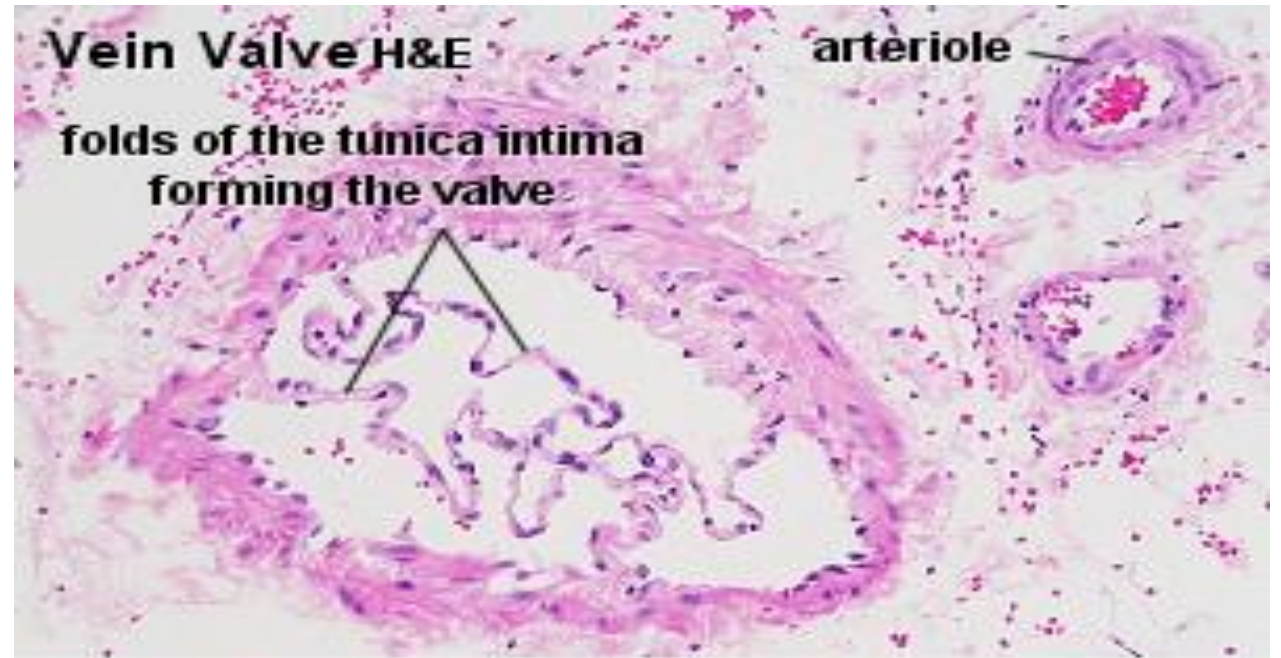
c. The tunica **adventitia is thicker** than the media 70% of the wall and frequently contains **longitudinal bundles of smooth muscle which strengthen the wall and prevent distention of the vessel.**

Both the media and adventitia contain elastic fibers

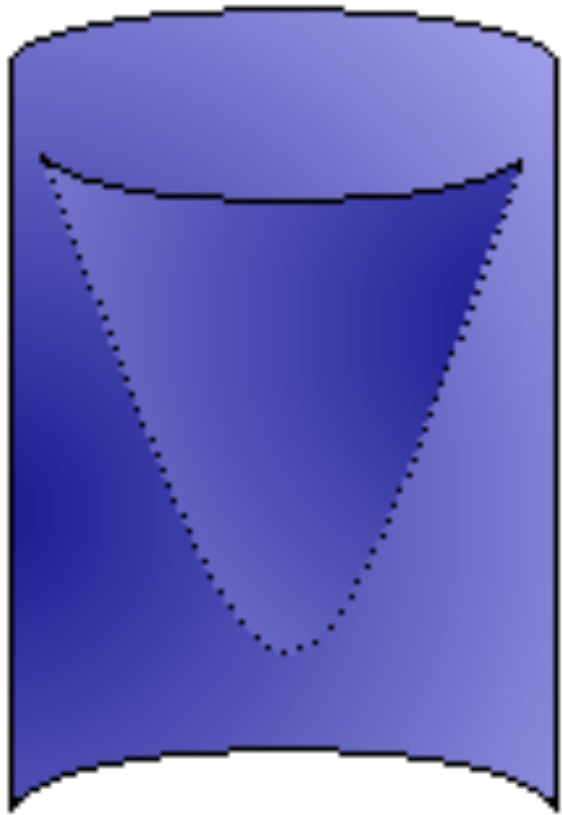


Vein valves

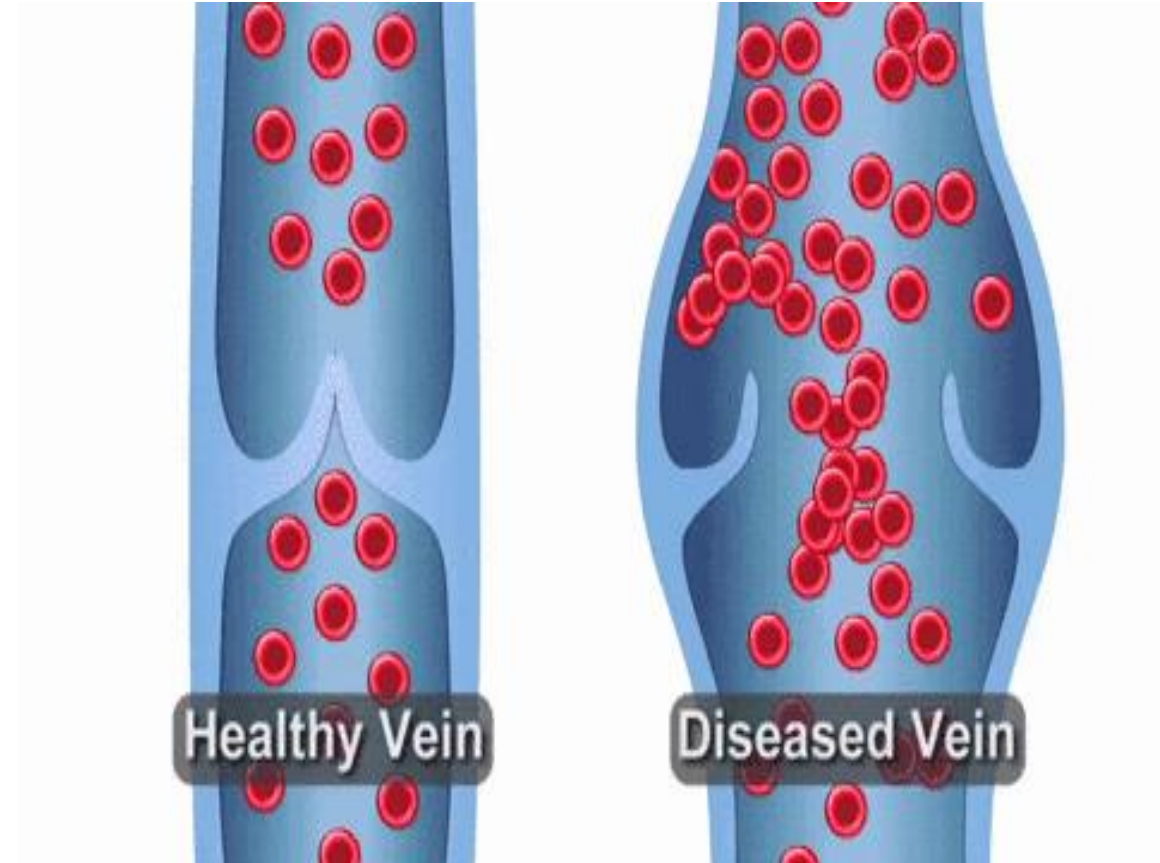
Which consist of thin, paired folds of the tunica intima projecting across the lumen, rich in elastic fibers and covered on both sides by endothelium

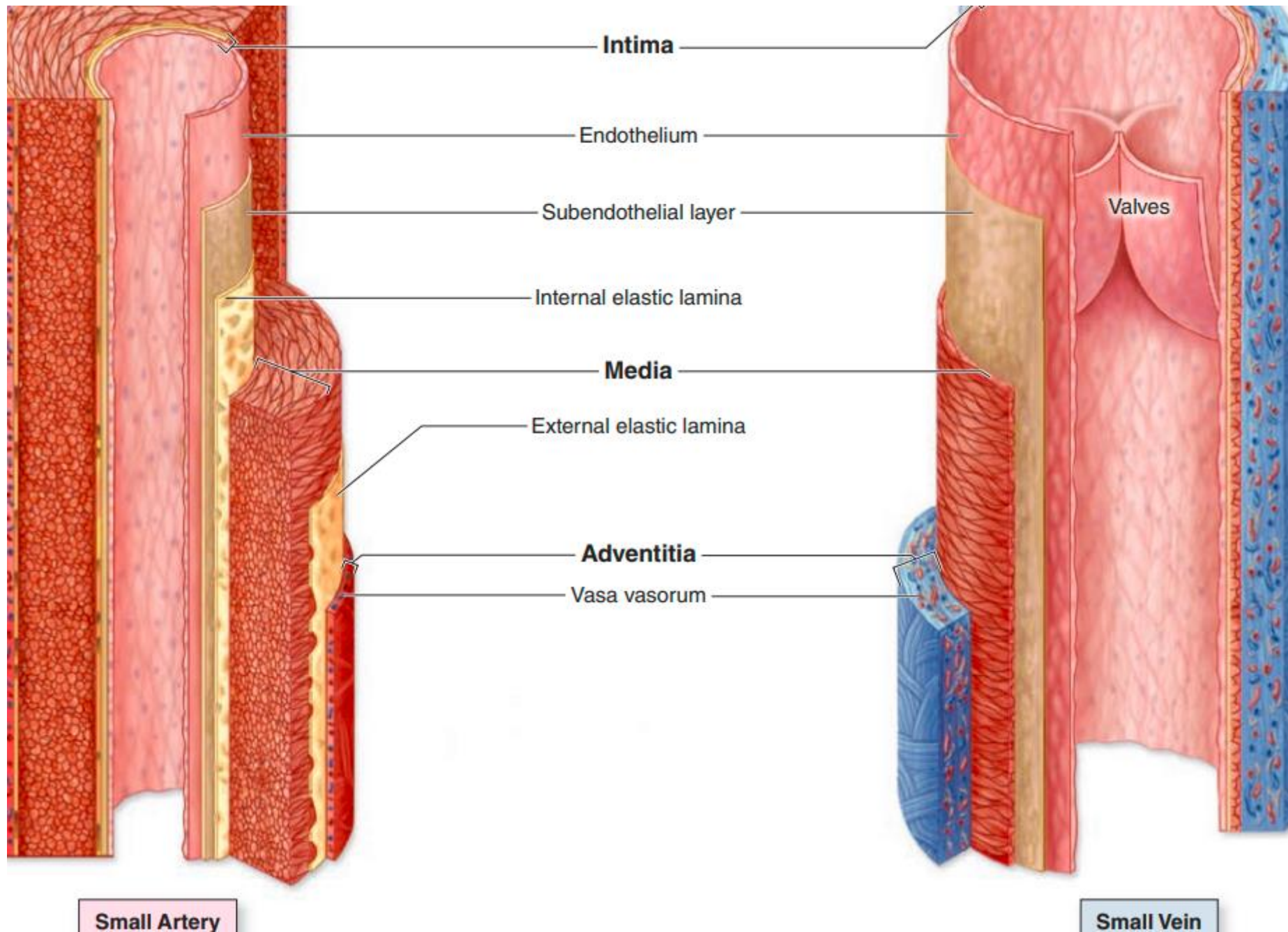


Vein valve



pocket valve





Size ranges, major features, and important roles of major blood vessel types.

Type of Artery	Outer Diameter (Approx. Range)	Intima	Media	Adventitia	Roles in Circulatory System
Elastic arteries	> 10 mm	Endothelium; connective tissue with smooth muscle	Many elastic lamellae alternating with smooth muscle	Connective tissue, thinner than media, with vasa vasorum	Conduct blood from heart and with elastic recoil help move blood forward under steady pressure
Muscular arteries	10-1 mm	Endothelium; connective tissue with smooth muscle, internal elastic lamina prominent	Many smooth muscle layers, with much less elastic material	Connective tissue, thinner than media; vasa vasorum maybe present	Distribute blood to all organs and maintain steady blood pressure and flow with vasodilation and constriction
Small arteries	1-0.1 mm	Endothelium; connective tissue less smooth muscle	3-10 layers of smooth muscle	Connective tissue, thinner than media; no vasa vasorum	Distribute blood to arterioles, adjusting flow with vasodilation and constriction
Arterioles	100-10 μ m	Endothelium; no connective tissue or smooth muscle	1-3 layers of smooth muscle	Very thin connective tissue layer	Resist and control blood flow to capillaries; major determinant of systemic blood pressure

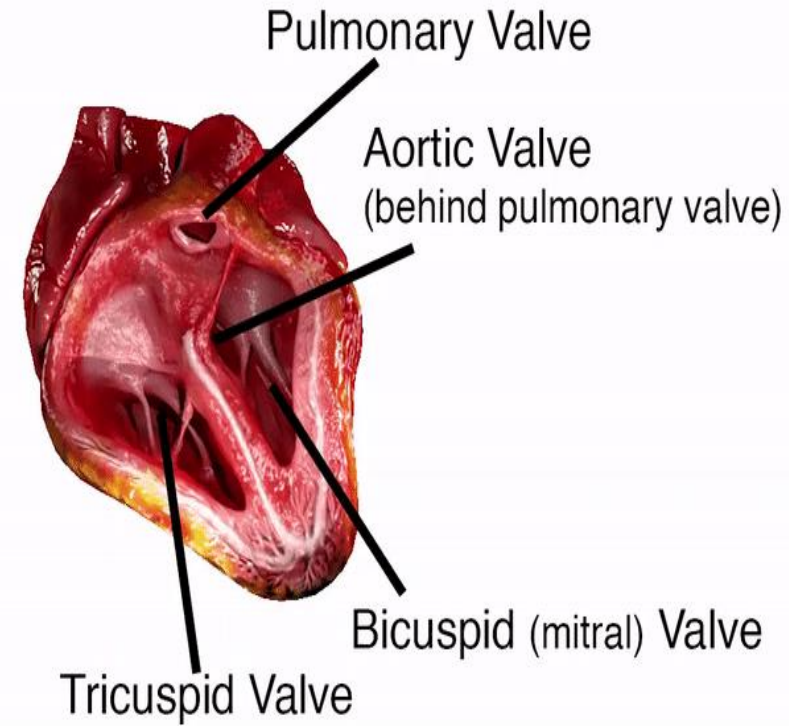
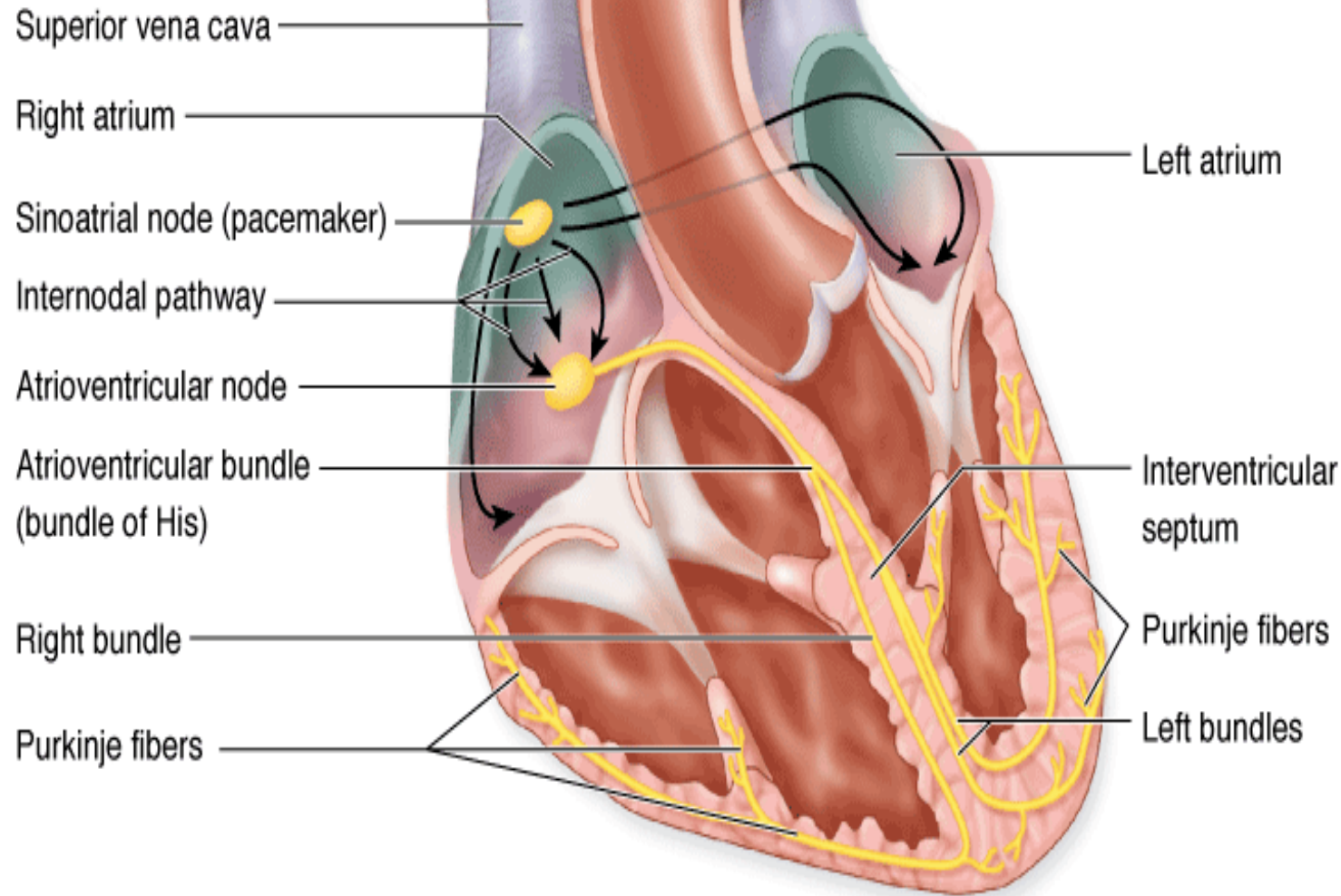


Size ranges, major features, and important roles of major blood vessel types.

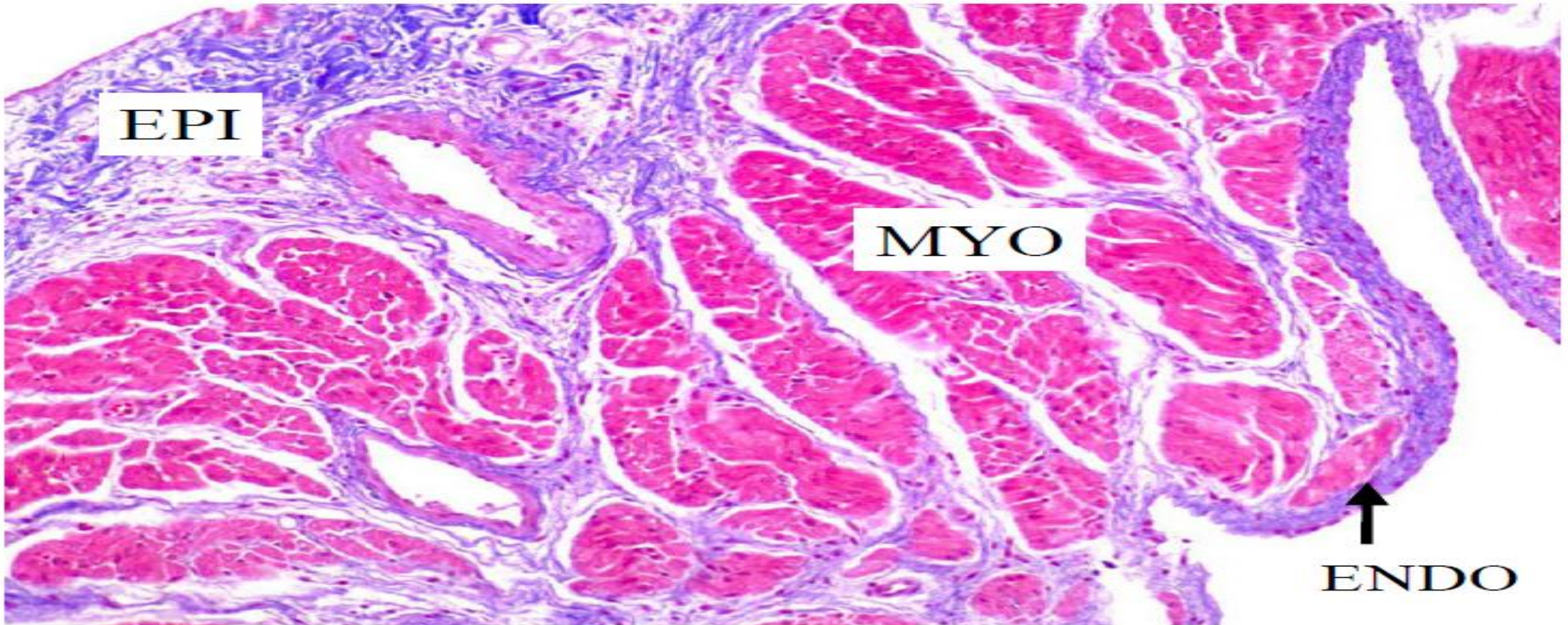
Capillaries	10-4 μm	Endothelium only	A few pericytes only	None	Exchange metabolites by diffusion to and from cells
Venules (postcapillary, collecting, and muscular)	10-100 μm	Endothelium; no valves	Pericytes and scattered smooth muscle cells	None	Drain capillary beds; site of leukocyte exit from vasculature
Small veins	0.1-1 mm	Endothelium; connective tissue with scattered smooth muscle fibers	Thin, 2-3 loose layers of smooth muscle cells	Connective tissue, thicker than media	Collect blood from venules
Medium veins	1-10 mm	Endothelium; connective tissue, with valves	3-5 more distinct layers of smooth muscle	Thicker than media; longitudinal smooth muscle may be present	Carry blood to larger veins, with no backflow
Large veins	> 10 mm	Endothelium; connective tissue, smooth muscle cells; prominent valves	> 5 layers of smooth muscle, with much collagen	Thickest layer, with bundled longitudinal smooth muscle	Return blood to heart



Heart



The Heart



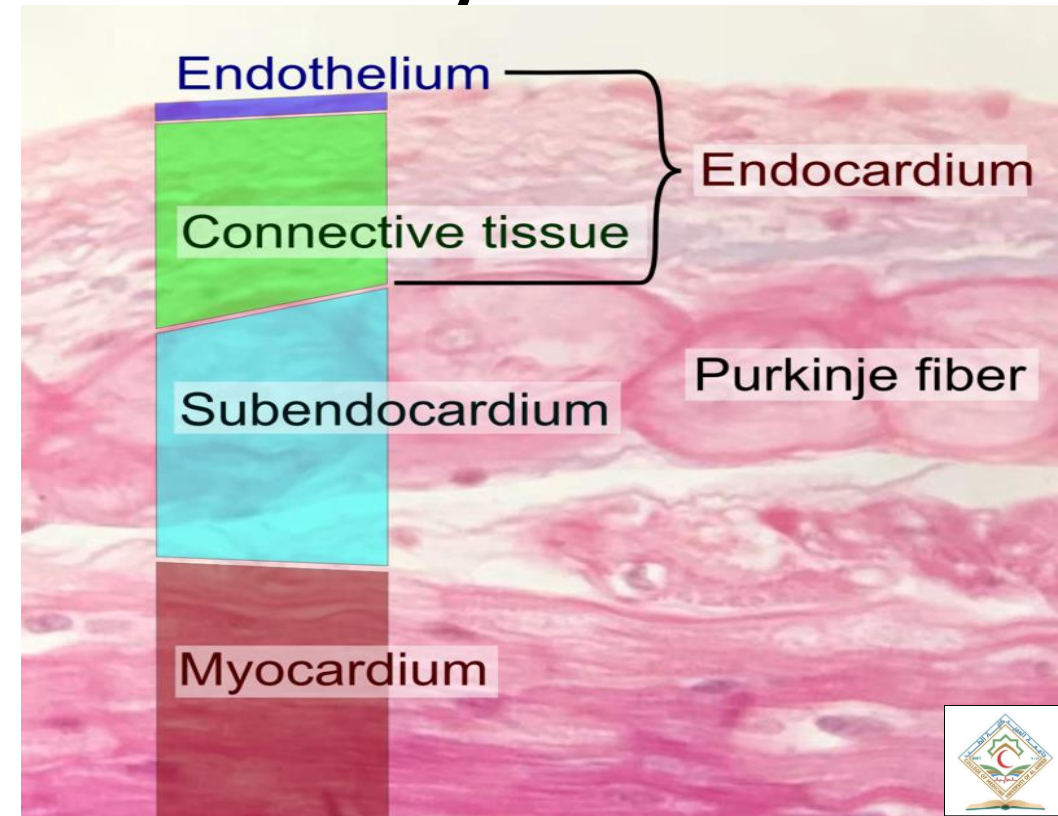
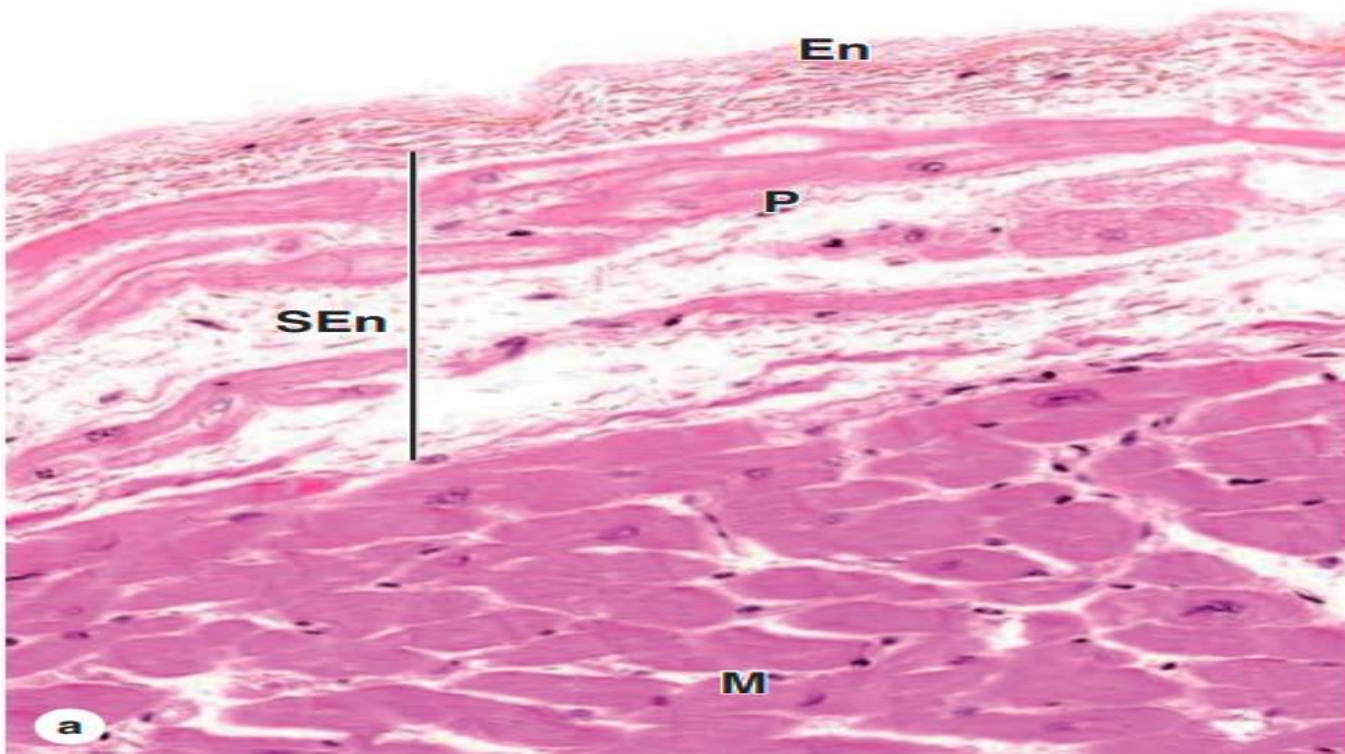
The walls of all four heart chambers consist of three major layers or tunics: the internal **endocardium**; the middle **myocardium**; and the external **epicardium**

1. Endocardium

A. Consists of a thin inner layer of **endothelium** and supporting fibroelastic connective tissue with scattered fibers of smooth muscle

B. Deep layer of connective tissue called the **subendocardial layer**.

Branches of the heart's **impulse-conducting system**, consisting of modified cardiac muscle fibers, are also located in the subendocardial layer .

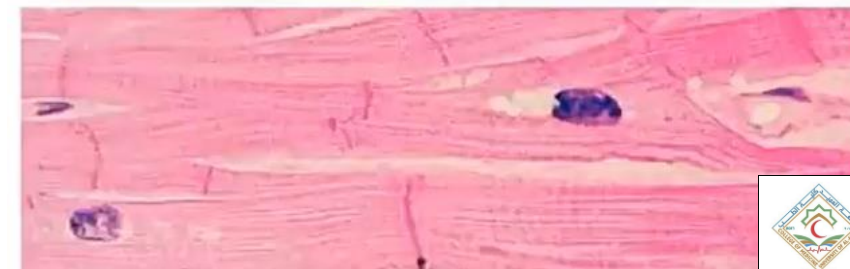
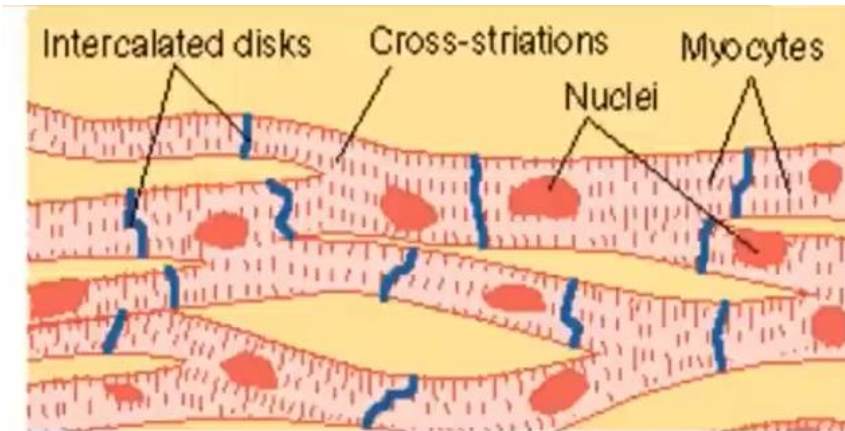
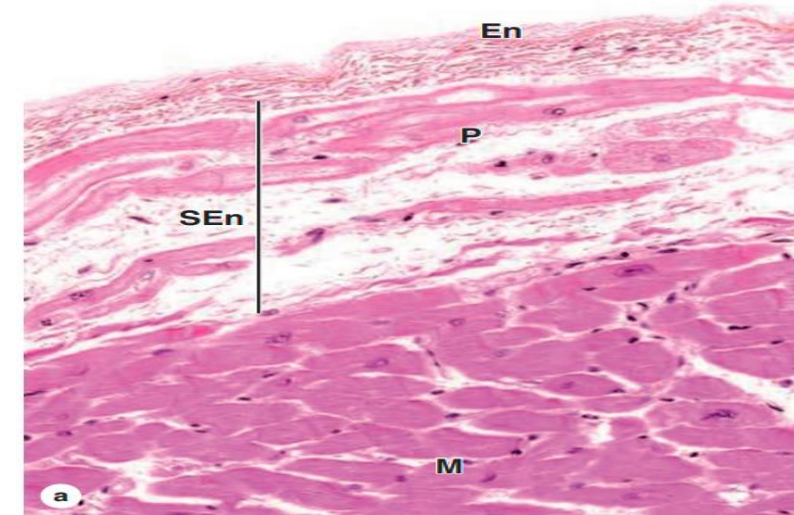


2. Myocardium

A. The thickest layer, consists mainly of cardiac muscle with its fibers arranged spirally around each heart chamber. Because strong force is required to pump blood through the systemic and pulmonary circulations, the myocardium is much thicker in the walls of the ventricles, particularly the left, than in the atrial walls .

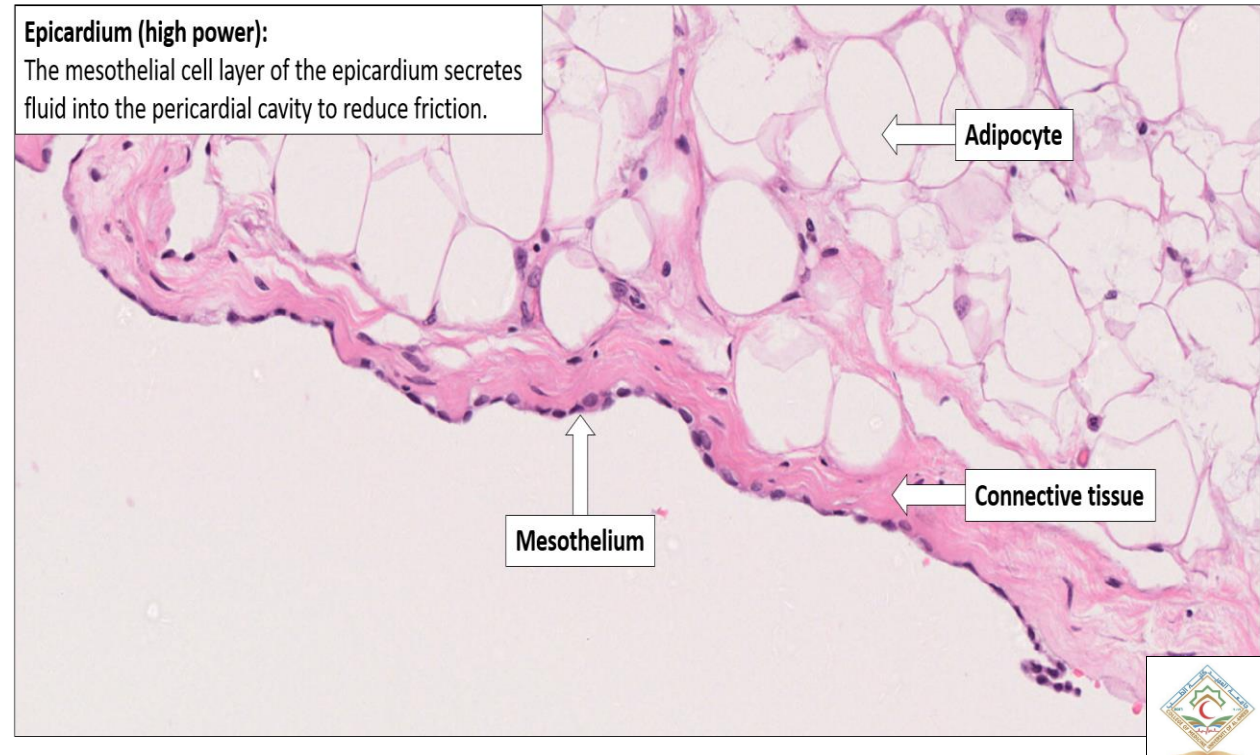
B. The cardiac muscle are long cylindrical fibers branch and anastomose. Show fainter striations than those of skeletal muscles. the cells attached by intercalated disks. The nuclei are single ,oval and centrally located.

C. The cardiac muscle fibers have no ability for regeneration . The injured cardiac muscles are replaced by fibrous tissue.



3. Epicardium

is a **simple squamous mesothelium** supported by a layer of loose connective tissue containing the coronary blood vessels, nerves and adipose tissue .The epicardium corresponds to the **visceral layer of the pericardium**.

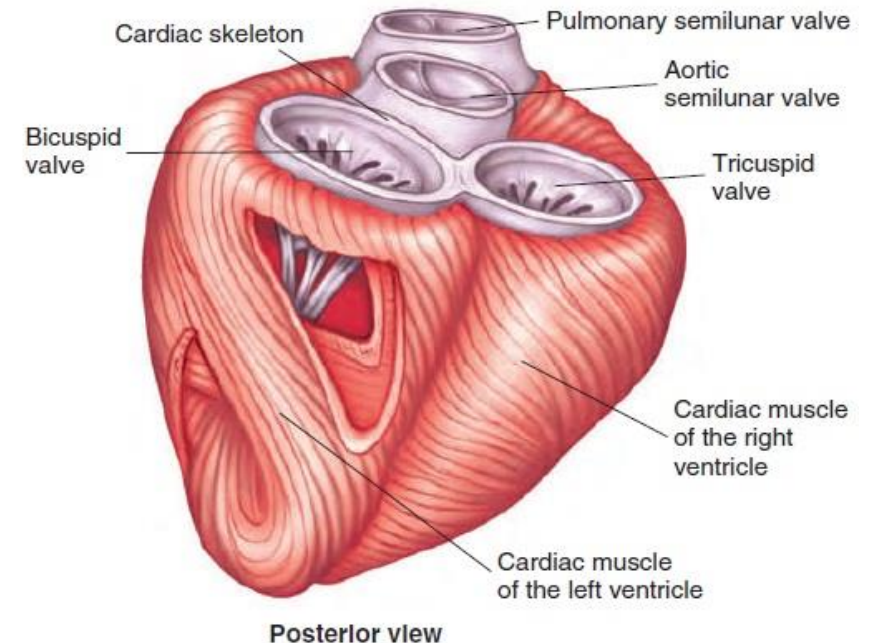
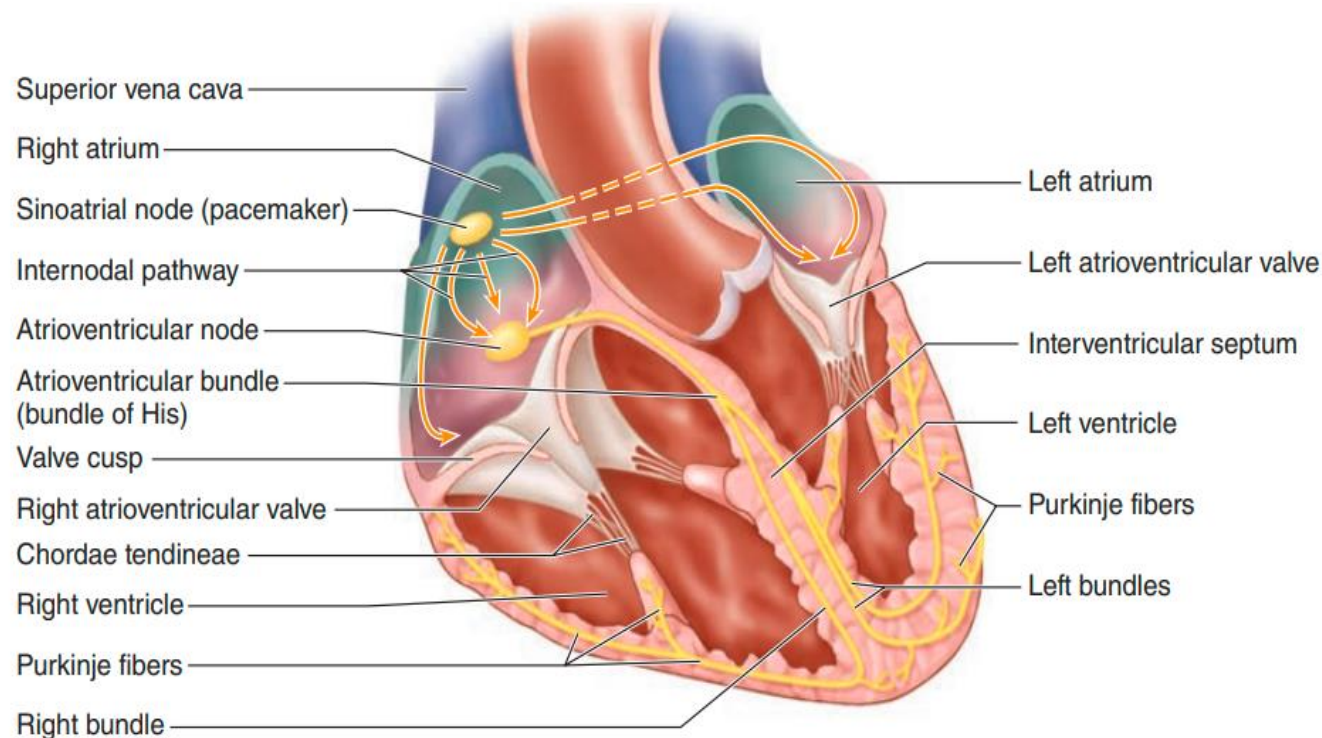


Cardiac skeleton

Prominent **dense irregular connective tissue** comprising the fibrous cardiac skeleton separates the musculature of the **atria from that of the ventricles**, forms part of the interventricular and interatrial septa, and extends into the valve cusps and the chordae tendineae to which they are attached.

functions:

1. Surrounding, anchoring, and supporting all heart valves
2. Providing firm points of insertion for cardiac muscle in the atria and ventricles
3. Helping coordinate the heart beat by acting as electrical insulation between atria and ventricles.

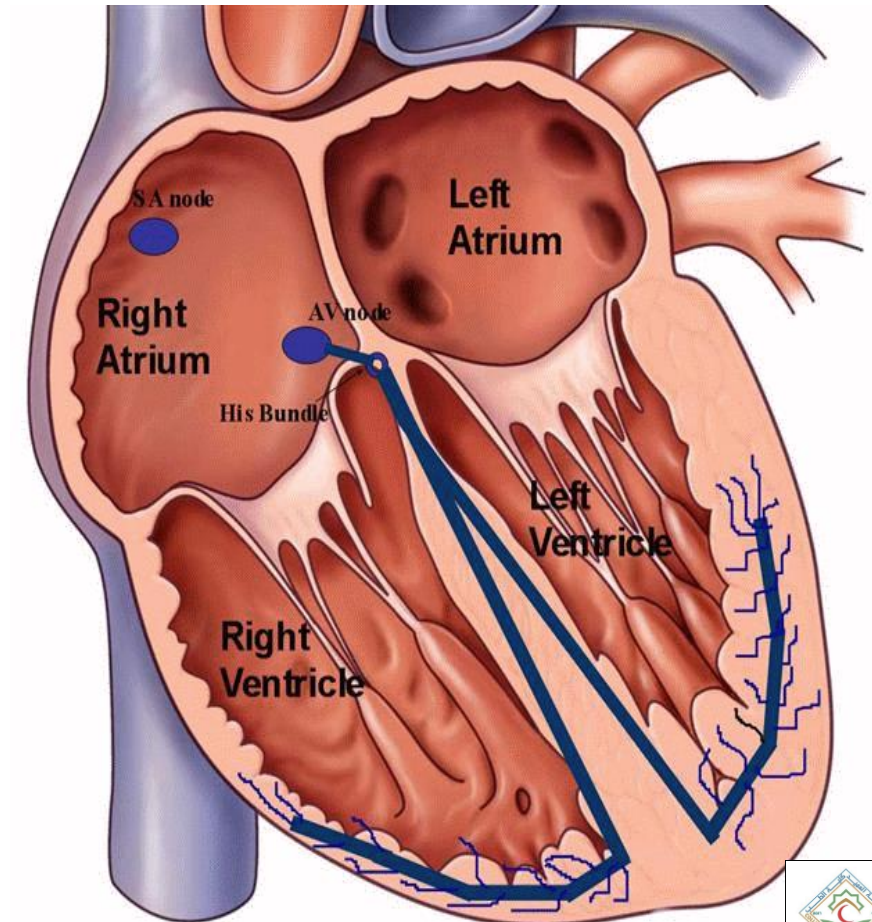
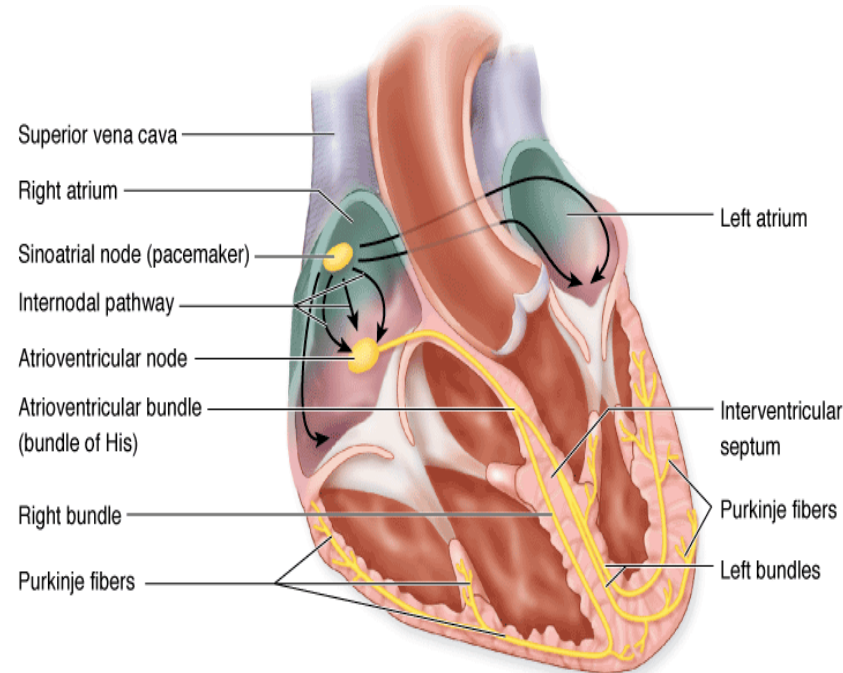


Conducting system of the heart

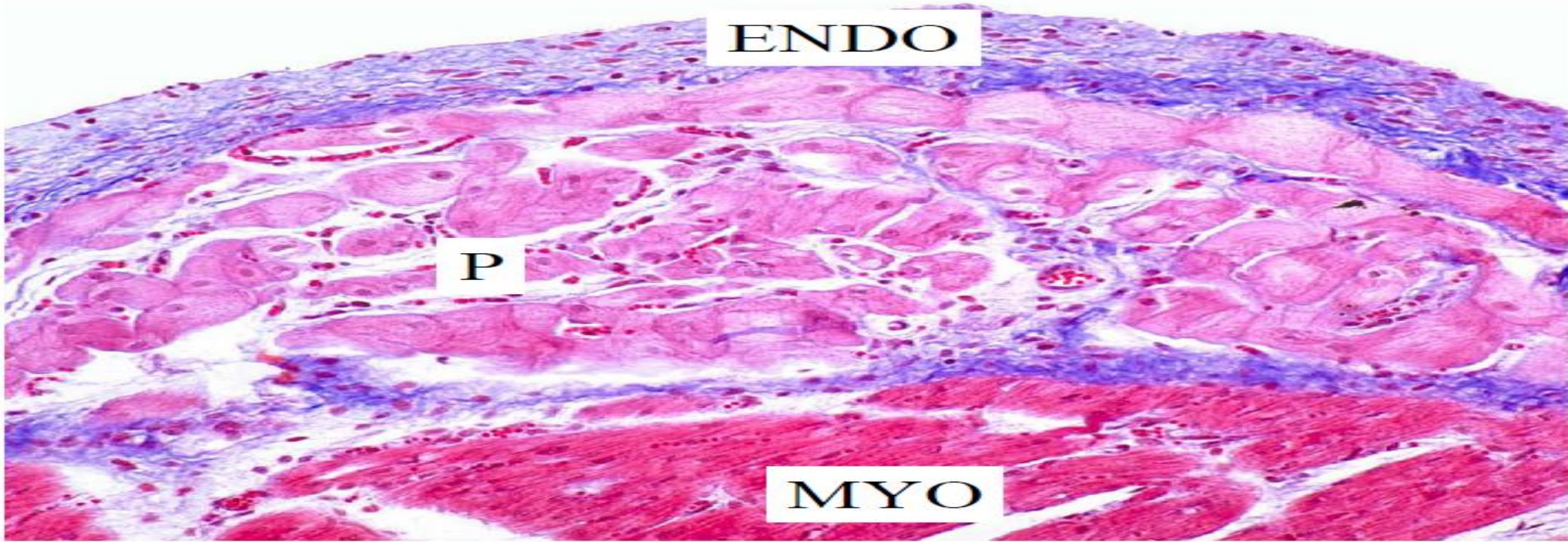
Modified cardiac muscle cells are specialized to generate and conduct waves of depolarization which stimulate rhythmic contractions in adjacent myocardial fibers. Present in both the **subendocardial layer and the adjacent myocardium**.

This system consists of two nodes of specialized myocardial tissue in the right atrial wall

- a. The sinoatrial (SA) node (or pacemaker)
- b. Atrioventricular (AV) node
- c. AV bundle (bundle of His) emerges from the (AV) node .



The Conducting System



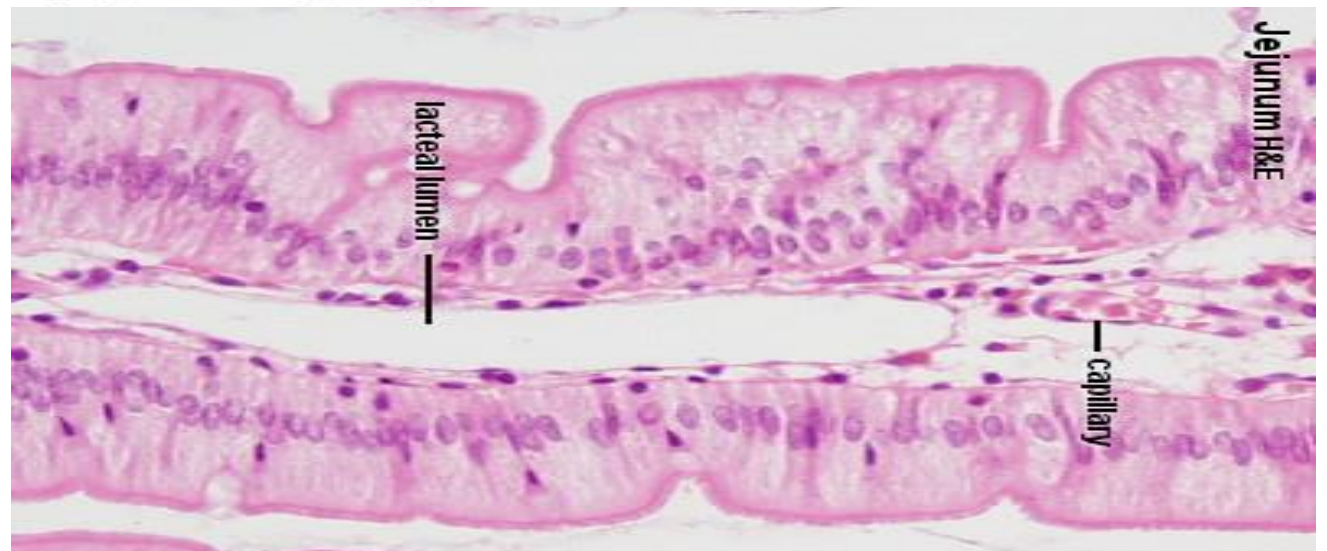
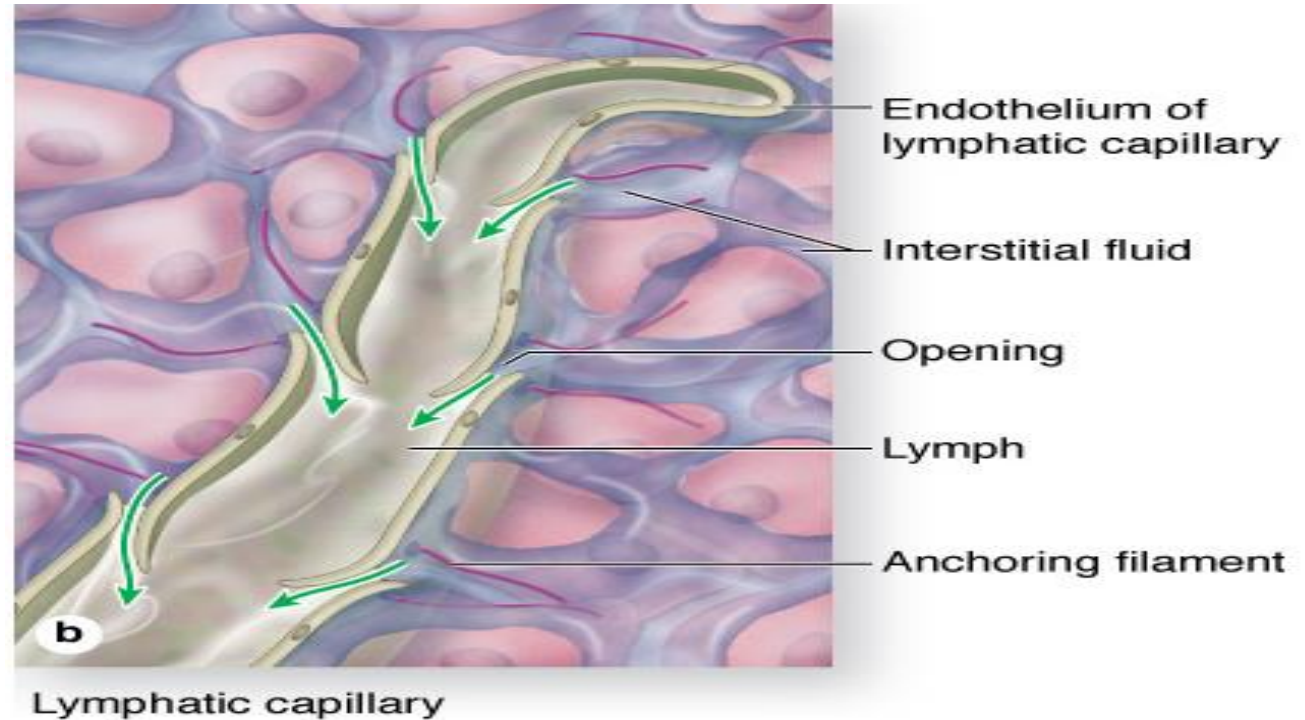
AV bundle, which passes through an opening in the cardiac skeleton into the interventricular septum where it bifurcates into left and right bundles located within the cardium and subendocardial layer. These conducting myofibers called **Purkinje fibers** are pale-staining fibers, larger than the adjacent contractile fibers, one or two central nuclei, myofibrils are sparse and restricted to the periphery of the cytoplasm with much glycogen



Lymphatic Vascular System

Lymphatic capillaries originate locally as tubes of very thin endothelial cells which **lack tight junctions and rest on a discontinuous basal lamina**. Fine anchoring filaments of collagen extend from the basal lamina to the surrounding connective tissue, preventing collapse of the vessels.

With exceptions such as the bone marrow and most of the CNS, most tissues with blood microvasculature also contain lymphatic capillaries (or lymphatics).



The larger lymphatic vessels have a structure **similar to that of veins** except that they have **thinner walls** and **lack a clear-cut separation** between tunics. They also **have more numerous internal valves**

Lymphatics and larger lymphatic vessels are clinically important because they facilitate the spread of pathogens, parasites, and malignant cells in the body. Surgical removal of lymph nodes, standard procedure to determine the occurrence of cancer metastasis

